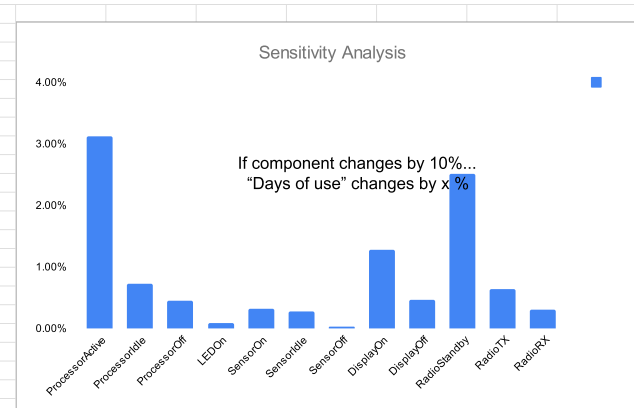


System Parameters (defined by hardware)		Profiles (usage of each component mode - defined by software and usage)		
form the datasheets		"off"	"sensing"	"interactive"
Processor				
Active	120 mW	0%	10%	50%
Idle	20 mW	0%	20%	30%
Sleep	1 mW	100%	70%	20%
LED				
On	5 mW	0%	5%	50%
Sensor				
On	10 mW	0%	10%	80%
Idle	2 mW	0%	90%	20%
Off	0 mW	100%	0%	0%
Display				
On	25 mW	0%	20%	100%
Off (leakage)	1 mW	100%	80%	0%
Radio				
Data Rate	300 bps	0%	0%	0%
Standby Power	5 mW	98%	95%	90%
TX Power	250 mW	0%	1%	5%
RX Power	120 mW	0%	1%	5%
Battery				
Capacity	500 mAh			
Nominal Voltage	3 V			
Regulator Efficiency	95%			
		16	7	1 hours/day typical usage



Total power in profile (mw)		Maximum Time
"off"	7.0009696 mW	223.9 hours
"sensing"	34.000764 mW	46.1 hours
"interactive"	125.100288 mW	12.5 hours
Effective Battery Capacity		
		1567.5 mW*h
Days of Use		3.30 days
Hours of Use		79.18 hours

REFLECTIONS : WHAT DID YOU LEARN FROM ANALYZING YOUR POWER. TALK ABOUT SOME POTENTIAL TRADEOFFS.

Days of use is calculated by combining the average daily energy consumption across off, sensing, and interactive modes based on expected daily usage hours.

I determined the days of use by estimating how the device would typically be used each day and dividing that time between off, sensing, and interactive modes. I then calculated the average daily energy consumption using the power values of each component and how often they are active in each mode. I chose the battery size by balancing how long the device should run with its size and cost. For the sensor device, a 500 mAh battery allows the device to operate for multiple days while staying small and lightweight. The display device uses more power because of the OLED screen and user interaction, so a larger 1000 mAh battery is more appropriate. To improve the user experience, power consumption could be reduced by using deeper sleep modes, sending data less frequently, lowering the LED brightness, or using BLE instead of Wi Fi. These changes would require more software effort and design decisions, but they would result in longer battery life and a more comfortable and usable device.

DATASHEETS	
ESP32	https://www.espressif.com/sites/default/files/documentation/esp32-c3_datasheet_en.pdf
LED — Standard	https://www.vishay.com/docs/83171/tlms1100.pdf
SENSOR - INMP441	https://invensense.tdk.com/wp-content/uploads/2015/02/INMP441.pdf
Display — SSD1306 OL	https://cdn-shop.adafruit.com/datasheets/SSD1306.pdf
Radio (ESP32C3)	https://www.espressif.com/sites/default/files/documentation/esp32-c3_technical_reference_manual_en.pdf
Batteries	https://www.adafruit.com/product/1578 https://www.adafruit.com/product/258